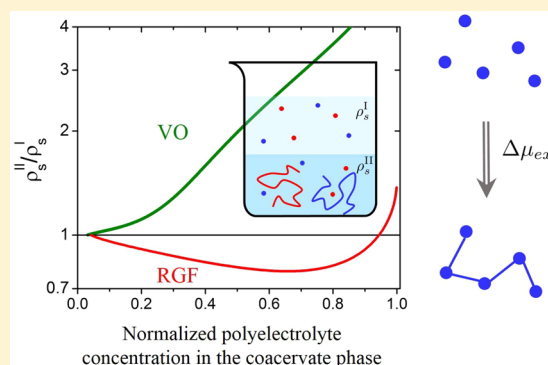


Salt Partitioning in Complex Coacervation of Symmetric Polyelectrolytes

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ABSTRACT: We perform a general thermodynamic analysis for the salt partitioning behavior in the coexisting phases for symmetric mixtures of polycation and polyanion solutions. We find that salt partitioning is determined by the competition between two factors involving the ratio of the polyelectrolyte concentration in the coacervate phase to that in the supernatant phase and the difference in the exchange excess chemical potential $\Delta\mu_{\text{ex}}$ —the excess chemical potential difference between PE segments and small ions—between the coexisting phases. The enrichment of salt ions in the coacervate phase predicted by the Voorn–Overbeek theory is shown to arise from its neglect of chain connectivity in the excess free energy which results in $\Delta\mu_{\text{ex}} = 0$ under all conditions. We argue that chain connectivity in general leads to a finite value of $\Delta\mu_{\text{ex}}$ which decreases with increasing PE concentration. Explicit calculations using theories that include the chain connectivity correlations—a simple liquid-state theory and a renormalized Gaussian fluctuation theory—show nonmonotonic behavior of the salt-partitioning coefficient (the ratio of salt ion concentration in the coacervate phase to that in the supernatant phase): it is larger than 1 at very low salt concentrations, reaches a minimum at some intermediate salt concentration, and approaches 1 at the critical point. This behavior is consistent with recent computer simulation and experimental results.



INTRODUCTION

When two solutions of oppositely charged polyelectrolyte (PE) are mixed together, under appropriate conditions the system can macroscopically separate into a supernatant phase that is depleted of PE chains and a coexisting liquid coacervate phase that contains most of PE chains as well as a large amount of water. This phenomenon is commonly termed complex coacervation.^{1,2} Because of its ubiquity in various industrial and biological systems, e.g., PE multilayer films^{3,4} and bioadhesives,^{5–7} complex coacervation has been extensively studied for many decades⁸ and is receiving renewed attention in recent years;^{9–24} we refer to several recent reviews^{2,25,26} for more comprehensive literature on the subject. There are many factors governing the phase behavior of these systems, including the charge fractions, chain lengths, pH, salt concentration, and temperature. In particular, salt concentration is one of the most important factors that can be easily and systematically used to control the coacervation behavior. For symmetric mixtures, the qualitative effect of salt is generally understood: with increasing salt concentration, the PE concentration difference between the coexisting coacervate and supernatant phases shrinks, and phase separation is completely inhibited once the salt concentration exceeds some critical value. However, there is no general agreement on salt partitioning in the coexisting phases; i.e., of the

supernatant phase and the coacervate phase, which phase has a higher salt concentration?

Experimentally, although there are many studies on polyelectrolyte complex coacervation, only a few have actually measured the small ion concentrations in the coexisting phases. Voorn and Overbeek's initial study²⁷ (for one coexisting condition of gum arabic at pH 3.75 which is expected to have low charge fraction) found that the small ion concentration in the coacervate phase was slightly higher than that in the supernatant phase by about 10%, producing a tie line in the PE–salt concentration phase diagram, with a positive slope. In the work of Spruijt et al.,²⁸ the phase diagram was constructed by assuming equal salt concentration in the coexisting phases, so no information on the salt partitioning was given. A recent study by Li et al.,²⁹ however, showed that over most salt concentrations the small ion concentration in the supernatant phase is higher than that in the coacervate phase; i.e., the tie lines have negative slopes, although the partitioning behavior is reversed at very low salt concentrations.

On the theoretical side, the classic Voorn–Overbeek (VO) theory, which combines the Flory–Huggins mixing entropy with the Debye–Hückel (DH) theory for the electrostatic

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correlations,³⁰ predicted that the coacervate phase has a higher small-ion concentration, consistent with their own experiment.²⁷ The generalized VO (GVO) theory study by Larson and co-workers³¹ predicted qualitatively the same salt partitioning behavior as the original VO theory. A field theoretic simulation (FTS) study based on one particular parameter set appeared to support this observation;³² however, that study did not systematically investigate the salt partitioning behavior.

On the other hand, theoretical studies using the random phase approximation (RPA)^{33,34} predicted higher concentration of the small ions in the supernatant phase than in the coacervate phase over a large parameter space. This partition behavior was also observed in several recent studies using other theories, such as the PRISM theory,³⁵ the first-order thermodynamic perturbation liquid state (TPT1-LS) theory,³⁶ and the renormalized Gaussian fluctuation (RGF) theory.³⁷ Recent computer simulation results also supported this observation.^{21,29} At very low salt concentrations, both RPA³³ and RGF³⁷ theories predicted enrichment of the small ions in the coacervate phase relative to the supernatant phase, consistent with the recent experimental results by Li et al.²⁹

The physical reasons for the different salt partitioning behaviors remain unclear. For example, Voorn and Overbeek argued that the enrichment of small ions in the coacervate phase in the VO theory arises from the stronger electrostatic correlation in this phase.²⁷ Sing and co-workers, on the other hand, invoked the *larger* excluded volume of chains in the coacervate phase as being responsible for the depletion of small ions in this phase.^{20,21,35} These arguments are debatable since the electrostatic correlation and the excluded volume of the PE chains are both larger in the coacervate phase than the supernatant phase for all theories. The effects of excluded volume were addressed more explicitly in the recent work of Li et al.,²⁹ which showed that by artificially weakening the excluded volume in the simulation, salt ions become enriched in the coacervate phase. However, the GVO theory, which improves upon the VO by incorporating excluded volume effects on the electrostatic correlation free energy,³¹ predicted qualitatively similar salt partitioning behavior to the VO theory. A satisfactory explanation for the different salt partitioning behaviors in polyelectrolyte complex coacervation predicted by the different theories is still lacking.

In this study, we provide a general thermodynamic analysis of the salt partitioning behavior in the coexisting supernatant and coacervate phases of symmetric polycation and polyanion mixtures, with the goal to elucidate the free energy contributions that affect the tie line slope. Starting from the phase equilibria conditions, we find that the ratio of small ion concentrations in the coexisting phases can be expressed in terms of the ratio of the PE concentrations in the coexisting phases and the difference in the *exchange excess chemical potential* between the supernatant phase and the coacervate phase, where the exchange excess chemical potential is defined as the difference in the excess chemical potential between the PE segments and the small ions. In the VO theory, this exchange chemical potential is identically zero; the theory therefore predicts that the coacervate phase *always* has a higher small ion concentration than the supernatant phase. On the other hand, the chain connectivity in the PE molecules leads to a finite exchange excess chemical potential, which decreases with increasing PE concentration; the resulting salt partitioning is a consequence of the competition between the exchange

excess chemical potential difference in the coexisting phases and the ratio of the PE concentration in the coexisting phases. Explicit calculations using the TPT1-LS theory^{36,38,39} and the RGF theory^{37,40} that incorporate this exchange chemical potential by accounting for the chain connectivity show depletion of small ions in the coacervate phase relative to the supernatant phase over a wide range of salt concentrations; we expect that the results will be qualitatively the same for other theories that account for chain connectivity.^{35,41–43} Interestingly, our calculations also reveal that for sufficiently low concentration of small ions the partitioning behavior is reversed. These results are consistent with computer simulations^{21,29} and recent experimental observations.²⁹

■ THERMODYNAMIC ANALYSIS

General Consideration. We consider symmetric mixtures of polycation and polyanion with small ions. For simplicity, we assume that all PE chains are fully charged with each chain consisting of N monomers and all charged species (including the PE segments and small ions) are monovalent and have the same size. Because of symmetry, the concentration of polycation is the same as that of polyanion, and we further denote them by ρ_p , i.e., $\rho_{p+} = \rho_{p-} = \rho_p$. Likewise, the concentration of small cation is the same as that of the small anion, and both are denoted by ρ_s , i.e., $\rho_+ = \rho_- = \rho_s$. The mixture is thus essentially a two-component system. The system Helmholtz free energy density can generally be written as the sum of an ideal contribution and an excess contribution:

$$f = f_{\text{id}} + f_{\text{ex}} \quad (1)$$

The ideal part is known exactly and is given by

$$\beta f_{\text{id}} = \frac{2\rho_p}{N} \left[\ln \left(\frac{\rho_p}{N} \Lambda_p^3 \right) - 1 \right] + 2\rho_s (\ln \rho_s \Lambda_s^3 - 1) \quad (2)$$

where $\beta \equiv 1/k_B T$ with k_B the Boltzmann constant and T the absolute temperature; Λ_p and Λ_s are length scales arising from integrations over the momentum degrees of freedom and have no consequence in the phase behaviors.

The exact expression for the excess Helmholtz free energy f_{ex} is unknown, and various approximations have been proposed.^{18,20,24,27,31,33,35,37–39} While these theories emphasize the different effects and thus may give different predictions with regard to the salt partitioning, here we provide a general thermodynamic analysis without specifying any particular expression of f_{ex} . Because of the intrinsic symmetry of the system between oppositely charged species, the Galvani potential—the electrostatic potential difference between the coexisting phases—is identically 0. Two-phase coexistence requires equality of the osmotic pressure and equality of the chemical potential of each charged species between these two phases. For definiteness, hereafter we denote the supernatant phase as phase I and the coacervate phase as phase II, and the concentrations of the species i in the supernatant phase and the coacervate phase are thus ρ_i^{I} and ρ_i^{II} , respectively. In particular, by splitting the chemical potential into an ideal contribution and an excess contribution, the equality of the chemical potential of polycations can be written in the form

$$\left(\frac{\rho_{p+}^{\text{II}}}{\rho_{p+}^{\text{I}}} \right)^{1/N} = \exp[\beta(\mu_{p+, \text{ex}}^{\text{I}} - \mu_{p+, \text{ex}}^{\text{II}})] \quad (3)$$

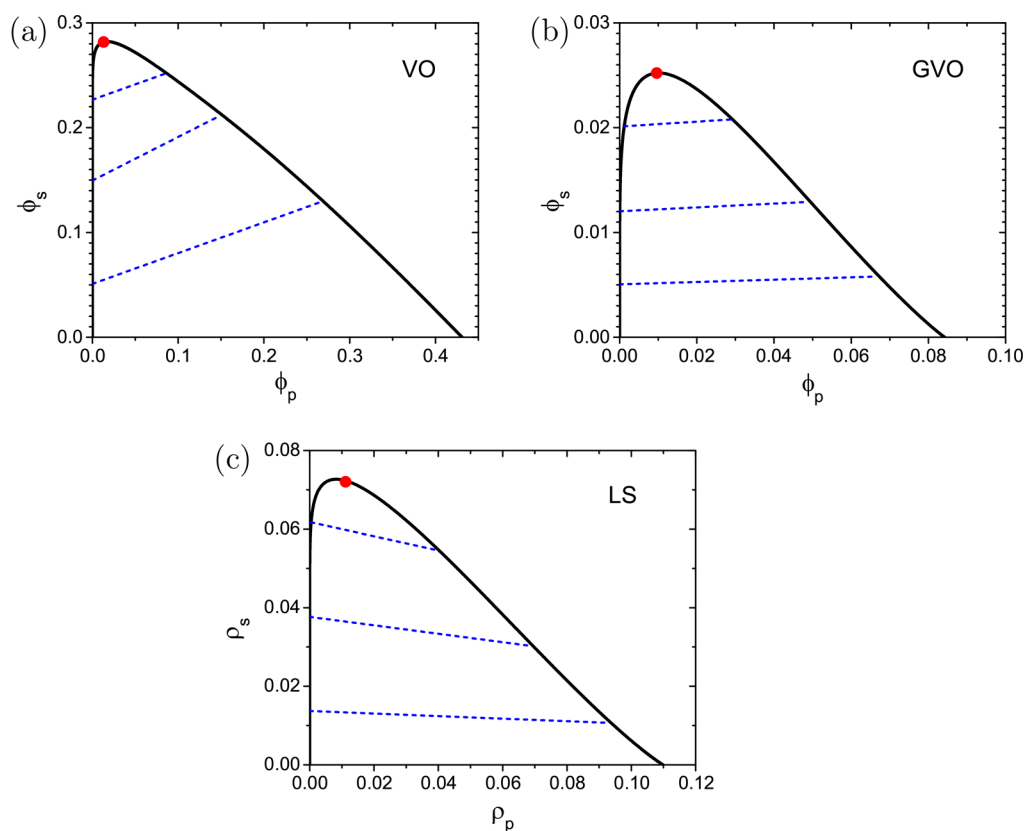


Figure 1. PE–salt concentration phase diagram calculated by (a) the VO theory, (b) the generalized VO theory, and (c) the TPT1-LS theory for systems with $l_B/\sigma = 1.785$ and $N = 100$. In the VO and GVO theory, $\phi_i \equiv \rho_i v$, which can be crudely related to the concentration ρ_i in the TPT1-LS theory by $\phi_i = (\pi/6)\rho_i\sigma_i^3$.

where $\mu_{p+,ex}^I$ and $\mu_{p+,ex}^{II}$ are the excess chemical potential per polycation segment in the supernatant phase and the coacervate phase, respectively. Likewise, the equality of the chemical potential of small cations in the coexisting phases reads

$$\frac{\rho_+^{II}}{\rho_+^I} = \exp[\beta(\mu_{+,ex}^I - \mu_{+,ex}^{II})] \quad (4)$$

where $\mu_{+,ex}^I$ and $\mu_{+,ex}^{II}$ are the excess chemical potentials of the cation in the supernatant phase and the coacervate phase, respectively. We note that by switching the “+” with the “−” in the subscripts, eqs 3 and 4 also hold true for polyanions and small anions; for concreteness, hereafter we only focus explicitly on the polycation and the small cation species when we talk about the PE and small ion species. Furthermore, for notational simplicity, hereafter we replace the subscript p+ and + by p and s, respectively.

From eqs 3 and 4, we have

$$\frac{\rho_s^{II}}{\rho_s^I} = \left(\frac{\rho_p^{II}}{\rho_p^I}\right)^{1/N} \exp[\beta\Delta\mu_{ex}^{II} - \beta\Delta\mu_{ex}^I] \quad (5)$$

where we have defined the exchange excess chemical potential $\Delta\mu_{ex}$ as the difference in the excess chemical potential between the polycation segment and the small cation, i.e., $\Delta\mu_{ex} \equiv \mu_{p+,ex} - \mu_{s+,ex}$. Physically, $2N\Delta\mu_{ex}$ represents the free energy change in removing N small cations and N small anions and simultaneously inserting one polycation and one polyanion or, equivalently, the free energy change in connecting N small

cations into one polycation and N small anions into one polyanion (note that both oppositely charged components need to be considered simultaneously to keep the system symmetric). Generally, $\Delta\mu_{ex}$ depends on both the PE concentration and the small ion concentration.

From eq 5, we see that the ratio of the small ion concentration in the coacervate phase and the supernatant phase, ρ_s^{II}/ρ_s^I , is determined by the ratio of the PE concentrations in the coexisting phases and the difference in the exchange excess chemical potential $\Delta\mu_{ex}$ between the coexisting phases. While in general ρ_p^I is smaller than ρ_p^{II} by many orders of magnitude, the $1/N$ power keeps $(\rho_p^{II}/\rho_p^I)^{1/N}$ in eq 5 not far from 1. Therefore, the sign and magnitude of the terms in the exponential factor play a key role in determining the salt partitioning behavior.

Analysis for the VO Theory. In the VO theory,^{17,27} the chain connectivity effect is completely neglected in the excess thermodynamic properties; the excess Helmholtz free energy density simply reads

$$\beta f_{ex} = \frac{1}{v}(1 - 2\rho_p v - 2\rho_s v) \ln(1 - 2\rho_p v - 2\rho_s v) - \frac{\kappa^3}{12\pi} \quad (6)$$

The first term in the above expression arises from the overall incompressibility for the PE species, small ions, and solvent, where v denotes the volume of a polymer segment or a small ion; this term effectively accounts for the excluded volume interaction between all charged species. The second term is the Debye–Hückel electrostatic correlation energy³⁰ for a dilute simple electrolyte solution, where $\kappa \equiv \sqrt{8\pi l_B(\rho_p + \rho_s)}$ is the

inverse of the Debye screening length, depending on the total concentration of all charged species, with l_B being the Bjerrum length, the distance at which the electrostatic interaction between two unit charges in the dielectric medium equals the thermodynamic energy $k_B T$. We stress that in the VO theory the chain connectivity effect is only accounted for in the translational degree of freedom but is completely neglected in the excluded volume interaction and the electrostatic correlation.

With eq 6, it is easy to see that the excess chemical potential of a polycation (or polyanion) segment is *exactly the same* as that of a small cation (or anion), i.e.

$$\beta\mu_{p,ex} = \beta\mu_{s,ex} = -\ln(1 - 2\rho_p v - 2\rho_s v) - 1 - \frac{\kappa l_B}{2} \quad (7)$$

Therefore, the difference of the excess chemical potential between the PE segment and the small ion vanishes under all conditions, i.e., $\Delta\mu_{ex} = 0$; eq 5 thus reduces to

$$\frac{\rho_s^{\text{II}}}{\rho_s^{\text{I}}} = \left(\frac{\rho_p^{\text{II}}}{\rho_p^{\text{I}}} \right)^{1/N} \quad (8)$$

Since $\rho_p^{\text{II}} > \rho_p^{\text{I}}$ by definition, we have $\rho_s^{\text{II}} > \rho_s^{\text{I}}$; i.e., the concentration of small ions in the coacervate phase is *always* higher than in the supernatant phase. This partitioning behavior was originally discovered by Voorn and Overbeek themselves numerically²⁷ and was also observed in more recent studies.^{17,21} In Figure 1a, we show the binodal as well as several representative tie lines in the PE–salt concentration phase diagram at $l_B/a = 1.785$ for systems with $N = 100$, where a here is some length scale related to the volume of a PE segment or small ion v . It is obvious that the slope of the tie line is positive. This figure serves as a reference for later comparisons with theories that include excluded volume in the electrostatic correlation and chain connectivity correlation.

It should be clear from these discussions that including excluded volume corrections to the electrostatic correlation energy, or having a different treatment of the excluded volume interactions than the incompressibility constraint, will not alter the qualitative behavior in salt partitioning, so long as the chain connectivity contribution to the excess free energy is ignored. As concrete examples, here we consider two alternative theoretical expressions for the excess free energy. The first replaces the DH electrostatic correlation (i.e., the second term of eq 6) in the VO theory by the generalized DH electrostatic energy, i.e., $\beta f_{\text{GDH}} = -[\ln(1 + \kappa a) - \kappa a + (\kappa a)^2/2]/4\pi v$; this expression has been used by Larson and co-workers recently and will be termed the generalized Voorn–Overbeek (GVO) theory hereafter.³¹ We note that the complete version of the GVO theory incorporates many other effects such as counterion condensation, ionic pairing, and Flory–Huggins interaction. However, to highlight the role of the excluded volume in the electrostatic correlation, here we only keep the generalized DH term as the driving force for complex coacervation in the GVO theory. The second approach is to model the ions and the charged monomers as charged hard spheres. For this model, the hard-core excluded volume interaction is described by the Carnahan–Starling expression^{44,45} i.e., $\beta f_{\text{hs}} = 6\eta^2(4 - 3\eta)/[\pi\sigma^3(1 - \eta)^2]$ where the packing fraction $\eta \equiv (\pi\sigma^3/6)\sum_{i=p,\pm} \rho_i$ with σ the diameter of the charged spheres, and the electrostatic correlation free energy in eq 6 is treated using the mean-spherical

approximation (MSA),^{46,47} i.e., $\beta f_{\text{MSA}} = -\Gamma^3(2/3 + \Gamma\sigma)/\pi$ where $\Gamma = (\sqrt{1 + 2\kappa\sigma} - 1)/2\sigma$ is the screening parameter. We shall refer to this theory simply as the MSA-VO theory hereafter. Note that both f_{GDH} and f_{MSA} reduce to the DH expression in the point charge limit.

On the basis of these two free energy expressions, we construct the binodal in the PE–salt concentration phase diagram. In Figure 1b, we show the ρ_p – ρ_s phase diagram at the same $l_B/a = 1.785$, calculated from the GVO theory; a similar figure is obtained from the MSA-VO theory and is not shown. Compared to the phase diagram obtained from the VO theory at the same l_B (i.e., Figure 1a), we see that the phase-separated region shrinks considerably due to the weaker electrostatic correlation. However, the tie lines in both cases have positive slopes, indicating that the coacervate phase has a higher small-ion concentration than the supernatant phase.

Our analyses thus demonstrate that the origin of the positive slopes in the tie lines in the PE–salt concentration phase diagram in the VO lies in the neglect of chain connectivity effects in electrostatic correlations, which results in $\Delta\mu_{ex} = 0$ under all conditions. GVO and MSA-VO, even with the explicit inclusion of excluded volume, share the same physical assumptions regarding chain connectivity and similarly end up with $\Delta\mu_{ex} = 0$. Therefore, all these theories predict eq 8.

In our analyses, we have taken the original and most common form of the VO theory in which the small-ion sizes have the same size as the monomer, all the monomers are charged, and the valencies of the charges of the small ions and monomers are the same. These effects generally results in $\Delta\mu_{ex} \neq 0$. It is straightforward to include these additional effects in the VO theory. We will not present the details of the analysis here as this would distract from the main purpose of this article. The result is that the exponential factor in eq 5 is less than 1 only when the charged monomer has higher valency than the small ions and/or the monomer size is smaller than the small-ion size, both unlikely to exist in experimentally relevant systems. For all other case, the exponential factor in eq 5 is larger than 1, and therefore the qualitative salt partitioning behavior is unaltered.

Our discussions so far have been restricted to systems with no other energetic interactions. Flory–Huggins interactions between the different species can have significant effects on the phase behavior in polyelectrolyte complex coacervation. For the symmetric mixture studied here, it is reasonable to assume the primary FH interactions to be between the polymer (both polycation and polyanion) and solvent, χ_{p0} , and between the polycation and polyanion, χ_{pp} . At the mean-field level, the additional excess free energy density is $f_{\chi} = k_B T[\chi_{p0}(\phi_{p+} + \phi_{p-})\phi_0 + \chi_{pp}\phi_{p+}\phi_{p-}]$ with $\phi_0 \equiv 1 - 2\rho_p v - 2\rho_s v$ being the volume fraction of solvent. For general χ_{p0} and χ_{pp} , the phase behaviors can become quite complex, including the possibility of microphase separation; we defer a systematic study to future work. Here we briefly discuss the qualitative effects of small to moderate χ_{p0} and χ_{pp} (such that other transitions do not interfere with the simple coacervation transition) on the salt partitioning behavior.

The FH interaction free energy f_{χ} results in the factor $\exp[\chi_{p0}(\phi_0^{\text{II}} - \phi_0^{\text{I}}) + \chi_{pp}(\phi_p^{\text{II}} - \phi_p^{\text{I}})]$ for the exponential term in eq 5. Noting that $\phi_p^{\text{II}} > \phi_p^{\text{I}}$ by definition, and $\phi_0^{\text{II}} < \phi_0^{\text{I}}$ in general,⁴⁸ we can conclude that increasing χ_{pp} at fixed χ_{p0} always increases the salt concentration in the coacervate phase, but increasing χ_{p0} at fixed χ_{pp} reduces the ratio $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ and can even make it less than 1 in some parameter space; explicit

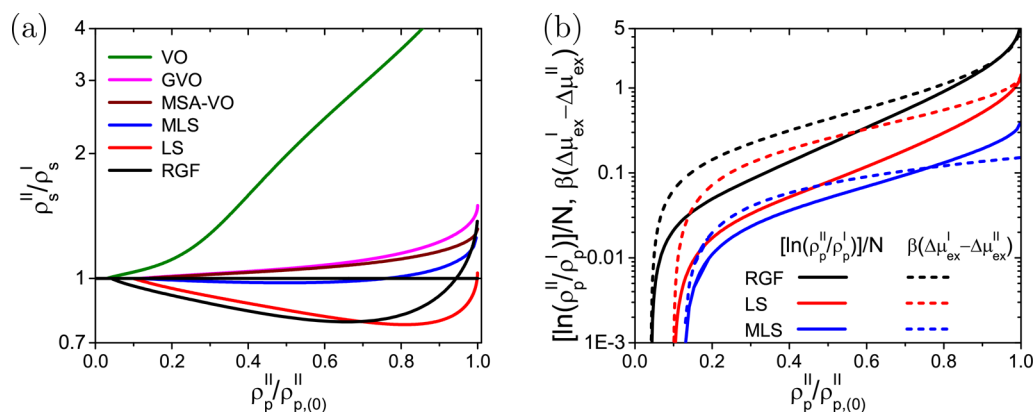


Figure 2. (a) Ratio of the small ion concentration in the coacervate phase to that in the supernatant phase $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ vs the PE concentration in the coacervate phase normalized by its salt-free value $\rho_{p+}^{\text{II}}/\rho_{p+,(0)}^{\text{II}}$ from various theories. (b) Two competing contributions in eq 5: $\ln(\rho_p^{\text{II}}/\rho_p^{\text{I}})/N$ and $\Delta\mu_{\text{ex}}^{\text{I}} - \Delta\mu_{\text{ex}}^{\text{II}}$ vs $\rho_{p+}^{\text{II}}/\rho_{p+,(0)}^{\text{II}}$ for the LS, MLS, and RGF theories. $l_B/\sigma = 1.785$ and $N = 100$. As a reminder, the acronyms in the legends are VO = Voorn–Overbeek, GVO = generalized Voorn–Overbeek, MSA-VO = mean-spherical approximation Voorn–Overbeek, LS = liquid state theory, MLS = modified liquid state, and RGF = renormalized Gaussian fluctuation.

calculations using the VO and GVO theories confirm these conclusions (data not shown). However, the magnitude of the reduction in $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ due to χ_{p0} is less than that due to the chain connectivity contribution to the exchange excess chemical potential to be discussed below. We note that our conclusions on the χ effects are quite general and are valid for any theory that treats the FH interactions at the mean-field level.

Effect of the Exchange Excess Chemical Potential.

Having identified the origin of the positive slopes of the tie lines in the VO and related theories, we now consider the properties of the exchange excess chemical potential between PE segments and small ions $\Delta\mu_{\text{ex}}$ (see eq 5) and examine how it affects the small-ion partitioning behavior. The excess chemical potential of each charged species can generally be decomposed into a term due to the chain connectivity and a term independent of the chain connectivity. As in the VO theory, the term independent of the chain connectivity is the same for the PE segments and for small ions and therefore does not contribute to $\Delta\mu_{\text{ex}}$. (To highlight the effects of chain connectivity and simplify the analysis, here we assume the monomers to be of the same size and valency as the small ions, and ignore the FH interactions.) Therefore, the only relevant term is that due to chain connectivity.

In view of the fact that the excluded volume interaction and the electrostatic correlation are the two main contributions in the excess properties, $\Delta\mu_{\text{ex}}$ can likewise be written as the sum of an excluded volume interaction contribution $\Delta\mu_{\text{ex}}^{(\text{ev})}$ and an electrostatic correlation contribution $\Delta\mu_{\text{ex}}^{(\text{el})}$, i.e., $\Delta\mu_{\text{ex}} = \Delta\mu_{\text{ex}}^{(\text{ev})} + \Delta\mu_{\text{ex}}^{(\text{el})}$. We stress that both terms here are due to chain connectivity since $\Delta\mu_{\text{ex}} = 0$ in the absence of chain connectivity. The small-ion concentrations in the coexisting phases are generally on the same order of magnitude (see Figure 1); however, PE concentrations between the coacervate phase and the supernatant phase differ by many orders of magnitude. We can therefore ignore the ρ_s dependence when considering $\Delta\mu_{\text{ex}}^{\text{II}} - \Delta\mu_{\text{ex}}^{\text{I}}$ in eq 5.

Since the exchange chemical potential $\Delta\mu_{\text{ex}}$ physically represents the free energy change in connecting N small cations (anions) into a polycation (polyanion) chain, we expect the excluded volume interaction part $\Delta\mu_{\text{ex}}^{(\text{ev})}$ to be negative by noting that some extra space is made available for other chains (and small ions) by localizing N disconnected particles into a connected chain (or a connected object of any

shape). Furthermore, the magnitude of this free energy decrease is larger for a higher PE concentration, and thus $\Delta\mu_{\text{ex}}^{(\text{ev}),\text{II}} < \Delta\mu_{\text{ex}}^{(\text{ev}),\text{I}}$. On the other hand, we expect the electrostatic correlation part $\Delta\mu_{\text{ex}}^{(\text{el})}$ to be positive, since connecting N small cations (anions) into a polycation (polyanion) results in the localization of many charges of the same sign of a smaller spatial region. The magnitude of this electrostatic correlation increase, however, decreases with increasing PE concentration because of screening, and thus $\Delta\mu_{\text{ex}}^{(\text{el}),\text{II}} < \Delta\mu_{\text{ex}}^{(\text{el}),\text{I}}$. Both physical arguments are consistent with results from the LS theory.^{38,39} Therefore, we have $\Delta\mu_{\text{ex}}^{\text{II}} < \Delta\mu_{\text{ex}}^{\text{I}}$ and the second (exponential) term in eq 5 is less than 1; see the dashes in Figure 2b. The salt partitioning behavior is a result of the competition between this term and the first term of eq 5 (the $(\rho_p^{\text{II}}/\rho_p^{\text{I}})^{1/N}$ term, which is larger than 1 by definition).

As an example, Figure 1c shows the binodal in the PE–salt concentration phase diagram at fixed $l_B/\sigma = 1.785$, calculated from a simple liquid state (LS) theory, where the chain connectivity is taken into account via the first-order thermodynamic perturbation theory (TPT1). We refer interested readers to refs 38, 39, and 36 for the details of the free energy expression in this TPT1-LS theory. The negative slope of the tie lines (shown by the blue dashes) indicates that the small ion concentration is higher in the supernatant phase than in the coacervate phase, in contrast to the tie line behaviors obtained from the VO and GVO theories (i.e., Figures 1a and 1b, respectively). Since both the excluded volume part and the electrostatic correlation part in the chain connectivity contribution of the exchange excess chemical potential result in $\Delta\mu_{\text{ex}}^{\text{II}} - \Delta\mu_{\text{ex}}^{\text{I}} < 0$, to evaluate the relative importance of these two contributions, we also perform a calculation using a modified LS theory (termed as the MLS theory hereafter), in which we only retain the excluded volume part in the chain connectivity. We find qualitatively similar features of the binodal to the full LS theory, with negatively sloped tie lines for higher small-ion concentrations. At low small-ion concentrations, the slopes of the tie lines turn positive in both the LS and the MLS results; see Figure 2a and discussions below.

In Figure 2a, we compare the predictions on the salt partitioning $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ from several different theories. As the binodals and critical points from the different theories are

different, to compare the data in the same range over which $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ varies, we choose the abscissa as the PE concentration in the coacervate phase ρ_p^{II} normalized by its value in the salt-free limit $\rho_{p,(0)}^{\text{II}}$. Physically, decreasing $\rho_p^{\text{II}}/\rho_{p,(0)}^{\text{II}}$ from 1 to 0 corresponds to tracing the coacervate branch of the binodal (shown in Figures 1) from the right end point (the salt-free limit) to the critical point. As expected, the VO, GVO, and MSA-VO theories, which all ignore the differences in the electrostatic correlation and excluded volume between the PE segments and small ions, predict $\rho_s^{\text{II}}/\rho_s^{\text{I}} > 1$ and monotonic decrease with decreasing $\rho_p^{\text{II}}/\rho_{p,(0)}^{\text{II}}$, in accordance with eq 8. Further, the ratio $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ from the VO theory is the largest among all the theories, in large part because the DH expression (i.e., the point charge approximation) in the VO theory drastically overestimates the electrostatic correlations, in comparison to GVO and MSA-VO.

The red, blue, and black curves in Figure 2a are results from the TPT1-LS (LS) theory, the modified LS (MLS) theory, and the renormalized Gaussian fluctuation (RGF) theory, respectively. All these theories predict that $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ is larger than 1 for $\rho_p^{\text{II}}/\rho_{p,(0)}^{\text{II}} \lesssim 1$ but is less than 1 for most of the range of $\rho_p^{\text{II}}/\rho_{p,(0)}^{\text{II}}$, with a minimum at some intermediate salt concentration. Moreover, by comparing the calculations from the LS theory and the MLS theory, we see that while both the excluded volume interaction part and the electrostatic correlation part in the chain connectivity contribution are responsible for the depletion of the small ions in the coacervate phase, the effect is dominated by the latter. Note that the LS theory, by including stronger correlation energies relative to the MLS theory, has a wider phase window than the latter and hence has a larger $(\rho_p^{\text{II}}/\rho_p^{\text{I}})^{1/N}$ ratio, which would favor salt enrichment in the coacervate phase; however, the Boltzmann weight of the exchange chemical potential difference far outweighs this enrichment effect.

The result from the particular implementation of RGF in ref 37 (where the excluded volume interactions are accounted for at the Flory–Huggins level, and hence $\Delta\mu_{\text{ex}}^{(\text{ev})} = 0$) further supports this conclusion. The main features in the salt partitioning behavior captured by the LS, MLS, and RGF, including the positive slope of the tie line at low salt concentrations, the negative slope at intermediate and high salt concentrations, and the existence of the minimum in $\rho_s^{\text{II}}/\rho_s^{\text{I}}$, are all qualitatively consistent with the recent experimental findings by Li et al.²⁹

The nonmonotonic salt partitioning behavior can be rationalized by the competition between the $(\rho_p^{\text{II}}/\rho_p^{\text{I}})^{1/N}$ term and the $\exp[\beta\Delta\mu_{\text{ex}}^{\text{II}} - \beta\Delta\mu_{\text{ex}}^{\text{I}}]$ factor in eq 5. In Figure 2b, we plot $[\ln(\rho_p^{\text{II}}/\rho_p^{\text{I}})]/N$ and $\beta(\Delta\mu_{\text{ex}}^{\text{II}} - \Delta\mu_{\text{ex}}^{\text{I}})$ separately for the LS, MSL, and RGF theories as a function of $\rho_p^{\text{II}}/\rho_{p,(0)}^{\text{II}}$. Note that $[\ln(\rho_p^{\text{II}}/\rho_p^{\text{I}})]/N$ is simply the difference in the ideal part of the chemical potential of the PE chain (per monomer) between the coacervate and the supernatant phases. As can be seen from Figure 2b, both terms increase monotonically (from 0) as $\rho_p^{\text{II}}/\rho_{p,(0)}^{\text{II}}$ varies from 0 at the critical point to 1 at the salt-free limit, but the initial increase is faster for $\beta(\Delta\mu_{\text{ex}}^{\text{I}} - \Delta\mu_{\text{ex}}^{\text{II}})$ than for $[\ln(\rho_p^{\text{II}}/\rho_p^{\text{I}})]/N$, resulting in $\beta(\Delta\mu_{\text{ex}}^{\text{I}} - \Delta\mu_{\text{ex}}^{\text{II}}) > [\ln(\rho_p^{\text{II}}/\rho_p^{\text{I}})]/N$ in a wide range of the plot. In this range, salt is depleted from the coacervate phase relative to the supernatant phase. However, as $\rho_p^{\text{II}}/\rho_{p,(0)}^{\text{II}}$ approaches 1, $[\ln(\rho_p^{\text{II}}/\rho_p^{\text{I}})]/N$ takes a more accelerated upward turn, eventually overtaking $\beta(\Delta\mu_{\text{ex}}^{\text{I}} - \Delta\mu_{\text{ex}}^{\text{II}})$, resulting in enrichment of small ions in the coacervate phase at low salt concentrations.

SUMMARY

In summary, on the basis of a general thermodynamic analysis, we have examined the salt partitioning behavior in the complex coacervation in symmetric polycation and polyanion mixtures. Starting from the equality of chemical potential for each species at coexistence, we find that the small-ion concentration ratio between the coexisting phases $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ can be expressed in terms of the ratio of PE concentration between the coexisting phases $\rho_p^{\text{II}}/\rho_p^{\text{I}}$ and the difference in the exchange excess chemical potential $\Delta\mu_{\text{ex}}^{\text{II}} - \Delta\mu_{\text{ex}}^{\text{I}}$ between the coacervate phase and the supernatant phase.

By treating the PE chains as disconnected objects in its expressions for the excluded volume and electrostatic correlation, the VO theory and its variants completely ignore the intrinsic asymmetry in the excess chemical potential between the PE segments and small ions due to chain connectivity and hence amounts to taking $\Delta\mu_{\text{ex}} = 0$ under all conditions; this immediately yields $\rho_s^{\text{II}}/\rho_s^{\text{I}} = (\rho_p^{\text{II}}/\rho_p^{\text{I}})^{1/N} > 1$. Thus, the small-ion concentration in coacervate phase is always higher than in the supernatant phase, and the ratio $\rho_s^{\text{II}}/\rho_s^{\text{I}}$ decreases monotonically with decreasing degree of phase separation. This conclusion is different from existing explanations in the literature that attribute the positively sloped tie lines in the VO theory either to stronger electrostatic correlations in the coacervate phase²⁷ or to the inadequate treatment of excluded volume effects.^{20,21,35}

On the other hand, we argue on physical ground that the exchange excess chemical potential in the coacervate phase is less than that in the supernatant phase which is responsible for the opposite salt partitioning behavior to the VO prediction over a wide range of small-ion concentrations. This salt partitioning behavior has been corroborated by calculations with the LS, and MSL, and the RGF theories, all accounting for the chain connectivity effect to varying degrees. In addition, all these three theories predict positive slopes in the tie lines for very low salt concentrations. These features are consistent with the recent experimental results.²⁹ Moreover, by comparing calculations from the LS theory and the MLS theory (where only the excluded volume part is accounted for in the chain connectivity contribution while ignoring its contribution to the electrostatic correlation), we find the electrostatic correlation correction due to the chain connectivity is the dominant effect in determining the negative slopes of the tie lines.

Finally, we stress that the thermodynamic analysis on the salt partition behaviors in this paper is general and independent of any specific model (as long as the symmetry between the cationic and anionic species is preserved). The present work highlights the importance of the exchange excess chemical potential between the PE segments and the small ions in explaining the salt partitioning behavior in the complex coacervation in symmetric polycation and polyanion mixtures. But eq 5 can be used in reverse to determine the exchange excess chemical potential difference between the coexisting phases from the coexistence data (using computer simulation, for example). This information can be helpful in guiding and validating future theories for polyelectrolyte complex coacervation.

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Notes

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